X-Gaussian Splatting

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Abstract

Goal of the project: Review of the Radiative Gaussian Splitting paper (***Cai et al., 2024a***) and an additional review of the related
Main results: Two reviewed papers buildung upon 3DGS technology and expanding its use on respective specific reconstruction problems.
Personal key learnings: An in-depth introduction to the concept of Gaussian-based 3D rendering and its applications in X-ray imaging. A further review of the literature helped to understand how GS-inspired design can improve rendering performance in different fields.
Estimated working hours: 8
Project evaluation: 4
Number of Words: 1595

Goal

The goal of this project is to understand the novel *X-Gaussian* framework proposed by ***Cai et al.*** (***2024a***), which introduces a 3D Gaussian Splatting-based method for X-ray novel view synthesis and a suggested related work from the literature (***Cai et al., 2024b***), focusing on advancements in 3D scene representation using High Dynamic Range Novel View Synthesis.

Introduction

X-ray imaging is essential in medicine because it provides internal views of the body without invasive procedures. However, excessive exposure to X-rays can be harmful, making it important to minimize both the amount and duration of radiation. Creating detailed three-dimensional images, such as those used in CT scans, typically requires many X-ray projections from different angles, which increases the overall radiation dose to the patient.
Advanced computational methods like Neural Radiance Fields (NeRF) can reconstruct 3D images from fewer X-ray views, but these approaches are often limited by high computational demands and slow processing times. In ***Cai et al.*** (***2024a***), a faster technique called 3D Gaussian Splatting (3DGS), originally developed for color images, has been adapted for X-ray imaging. In this new form, known as Radiative Gaussian Splatting (X-Gaussian), the method takes into account how X-rays are absorbed by different tissues rather than how light reflects from surfaces. This adaptation enables fast and accurate 3D reconstruction from fewer X-ray images, helping to reduce radiation exposure while maintaining high image quality.

Methods

The process begins with multiple 2D X-ray images from different angles. Unlike classical 3D Gaussian Splatting (3DGS) (***Kerbl et al., 2023***), which uses color images and Structure-from-Motion (SfM) as its initial inputs, X-Gaussian relies solely on multi-view X-ray images to define a 3D cuboid via Angle-pose Cuboid Uniform Initialization (ACUI), based on known imaging geometry. Gaussian

splats are uniformly distributed within this cuboid, each representing view-independent tissue penetrability rather than view-dependent color or opacity.
During training, Differentiable Radiative Rasterization (DRR) simulates X-ray transmission through the Gaussian cloud, generating synthetic X-ray images that are compared to real ones. The model parameters, like position, size, and radiodensity of the Gaussian Splats, are optimized during each iteration. After training, the model can synthesize X-ray images from novel viewpoints and reconstruct high-quality 3D anatomy.
This approach enables much faster training and inference than NeRF-based models and achieves accurate 3D reconstructions from fewer X-ray projections, reducing patient exposure.

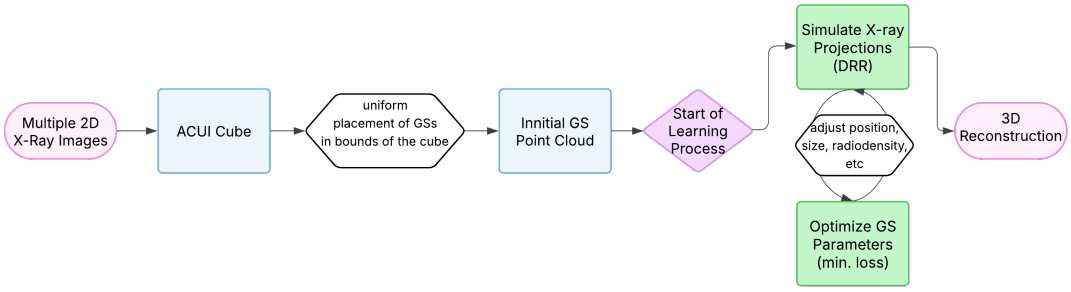


Figure 1. Graphical Abstract of the Radiative Gaussian Splatting pipeline for X-ray 3D reconstruction.

Experiments

The authors evaluate X-Gaussian using real CT datasets, generating 100 simulated X-ray projections per object from multiple angles. For novel view synthesis, 50 images are used for training and 50 for testing. X-Gaussian is compared against five leading NeRF-based models, with performance measured by reconstruction accuracy (per-pixel value accuracy (PSNR), overall structural similarity (SSIM), training time, and inference speed.
A sparse-view experiment tests the ability of the models to help reconstruct CT-scans from sparse data. Three standard CT-reconstruction methods are utilized to reconstruct a CT scan from 5 images alone and with the help of 95 novel images, X-Gaussian, and five other NeRF based methods each. These are evaluated on the same metrics as above.
Additional ablation studies assess the impact of model components, including different point cloud initialization strategies and the number and spacing of Gaussian splats. The authors also analyze convergence speed and investigate how the number of training images affects the covariance (size and shape) of the Gaussian splats.

Summary and Results

In the novel view synthesis experiment, X-Gaussian achieves higher PSNR and SSIM scores than all five NeRF-based baselines, while also providing much faster training and inference times. Specifically, the method achieves up to +6.5 dB improvement in PSNR, offers 73× faster inference, and requires only 15% of the training time compared to the recent best NeRF based method. Additionally, X-Gaussian provides a more detailed reconstruction with fewer artifacts.
In the sparse-view reconstruction task, X-Gaussian consistently outperforms both NeRF-based models and the use of traditional algorithms alone. The ablation study shows that removing or altering key components of the method, such as initialization or the number of Gaussians, leads to decreased reconstruction quality, highlighting the importance of parts like 3D cube initialization or radiodensity modeling to the pipeline.
Lastly, convergence analysis shows that X-Gaussian converges earlier. Covariance analysis shows that increasing the number of training views leads to finer and more precise 3D Gaussian point

clouds, improving the model's ability to represent detailed anatomical structures.
All in all, the proposed X-Gaussian method provides a faster and more reliable solution for X-ray image reconstruction, achieving higher quality and efficiency than state-of-the-art approaches. However, as noted in the paper, its complex technical implementation may make it harder to debug and adopt in practice. Overall, X-Gaussian represents a significant advancement with potential for improving clinical X-ray imaging.

Literature Review

A related work by the same author(s) has been done using Gaussian-based point-cloud representations accelerated by rasterization. The HDR-GS novel view synthesis ***Cai et al.*** (***2024b***) adapts for High Dynamic Range (HDR) imaging, integrating spherical harmonics that are used for HDR modeling and then tone mapping for consequent LDR modeling. This paper closely aligns with X-Gaussian, as it also builds on 3D Gaussian Splatting but applies it to HDR rendering, which is another specialized imaging task that benefits from GS-inspired rendering models and results in fast inference, hence the reason for choosing this article.

Introduction

The HDR-GS paper ***Cai et al.*** (***2024b***) builds on 3D Gaussian Splatting (3DGS) to enable fast, high-quality novel view synthesis for scenes with a wide high dynamic range (HDR)—that is, scenes containing both extremely bright and very dark regions that standard images cannot fully capture. Its key innovation is separating the representation of HDR radiance (using spherical harmonics per 3D Gaussian) from LDR image display (via a neural network that generates tone-mapped low dynamic range (LDR) images at arbitrary, user-controlled exposure settings). Trained on multi-view LDR images with known exposure times, HDR-GS learns to reconstruct a full HDR radiance field and render both HDR and LDR images. This design allows faithful reproduction of scene brightness and flexible exposure control in real time, while remaining significantly faster than prior NeRF-based methods and demonstrating the extensibility of 3DGS to advanced HDR imaging tasks.

Methods

Standard 3D Gaussian Splatting (3DGS) represents scenes using 3D Gaussians that store geometric properties such as position, scale, and opacity and encode view-dependent colors within the limited low dynamic range (LDR) of typical RGB images. This makes standard 3DGS inadequate for the rendering of scenes with significant variations in exposure or the wide spectrum of brightness found in high dynamic range (HDR) environments.
The HDR-GS framework overcomes these limitations by first applying Structure-from-Motion (SfM) to multi-view LDR images with known exposure times. This recalibrates camera parameters and initializes a 3D Gaussian point cloud in metric space. HDR-GS then introduces a Dual Dynamic Range (DDR) Gaussian model, where each Gaussian uses spherical harmonics (SH) to represent HDR radiance. To enable exposure control and robust tone-mapping, three neural networks (MLPs), one for each RGB channel, are trained to convert HDR color into LDR color based on user-specified exposure. Finally, HDR-GS employs Parallel Differentiable Rasterization (PDR), simultaneously rendering both HDR images (from SH coefficients) and exposure-adjustable LDR images (from the MLP tone-mapping outputs). During training, where the rendered and ground-truth images are compared, model parameters, including the position, scale, rotation, opacity, spherical harmonics coefficients, and MLP weights, are optimized.
This dual-pipeline design allows HDR-GS to faithfully reconstruct the full dynamic range of a scene and generate high-quality images at any chosen exposure, while retaining the speed and scalability advantages of 3DGS.

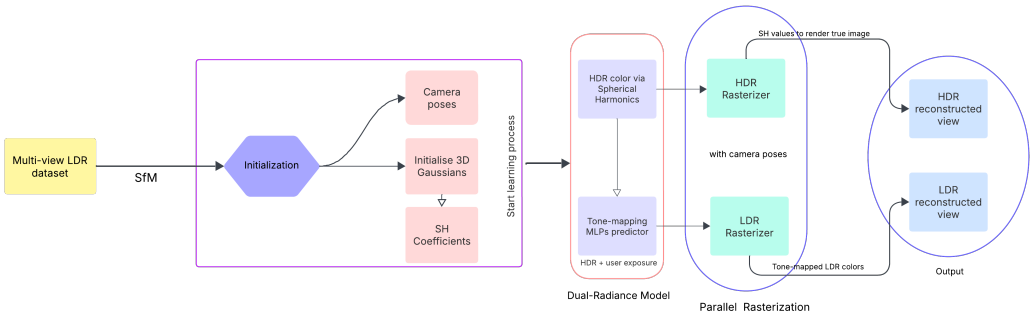


Figure 2. Graphical Abstract of the HDR-GS Pipeline for Novel HDR View Synthesis and User-Controlled LDR View Synthesis.

Experiments

The authors evaluated HDR-GS using the HDR-NeRF dataset, which contains 4 real-world and 8 synthetic scenes. Each scene was captured from 35 different camera viewpoints and photographed under five different exposure settings, ranging from dark to bright. For training, images were taken from 18 viewpoints using a subset of the available exposure settings. The remaining 17 viewpoints, along with both the training and intermediate exposure settings, as well as the true HDR images, were used for evaluation and testing.

For benchmarking, HDR-GS was compared with state-of-the-art methods, including HDR-NeRF and Instant-NGP, using the same image quality metrics as in X-Gaussian like PSNR, SSIM, and also LPIPS (Learned Perceptual Image Patch Similarity).

Additionally, the authors performed ablation studies, similarly to the X-Gaussian ablation studies, to systematically evaluate the impact of different components and design choices within the HDR-GS pipeline. These studies involved modifying or removing specific elements, such as the tone-mapping strategy and exposure settings during training, to assess their influence on overall performance.

Results and Summary

The proposed HDR-GS framework achieves state-of-the-art results in high dynamic range (HDR) novel view synthesis, significantly outperforming prior NeRF-based methods in both image quality and efficiency. Experiments show that HDR-GS improves PSNR by 3.84dB for LDR and 1.91dB for HDR synthesis compared to previous approaches, while uniquely enabling user-controllable exposure settings for flexible, real-time tone-mapped rendering of LDR images. Importantly, HDR-GS achieves up to 1000× faster inference and requires only 6.3% of the training time, demonstrating its practical advantages for applications demanding photorealism, dynamic lighting, and interactive performance.

Discussion

This article is important for future research because it introduces HDR-GS, a framework specifically designed for high dynamic range (HDR) imaging and user-controlled exposure rendering. By enabling real-time, photorealistic visualization with adjustable exposure, HDR-GS paves the way for new possibilities in AR/VR, digital content creation, and scientific imaging.

A potential application could lie in fluorescence microscopy, where biological tissues often contain both extremely bright and very faint structures, the ability to dynamically control exposure and visualize all details without saturation or loss is crucial for accurate analysis and diagnosis. Techniques like HDR-GS, which reconstruct HDR radiance from multi-exposure LDR images, could provide a valuable foundation for future advances in high-fidelity biomedical imaging.

As interactive and high-fidelity 3D rendering becomes increasingly important, the HDR-GS frame-

work offers a strong foundation for advances in both visual computing and biomedical research.

Contributions

1. Attempts in running the code and review of the Radiative Gaussian Splatting paper by Ksenia Krasikova.
2. Literature Review and discussion on HDR-GS novel synthesis by Payal Priyadarshini.
3. Philipp Harlos was excused (was absent).

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